MIL-B-44134
13 AUG 1984
(See 6.6)

MILITARY SPECIFICATION

BODY ARMOR, SMALL ARMS (HAND GUN) PROTECTIVE,

UNDERGARMENT, MP-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a small arms protective undergarment having removable front and back torso ballistic protective inserts.

1.2 Classification. The body armor shall be one type in the following sizes (see 6.2):

Size

Small Regular
Medium Regular
Large Regular
Extra Large Regular
Small Long
Medium Long
Large Long
Extra Large Long

2. APPLICABLE DOCUMENTS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8470

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2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label: for Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-F-21840	-	Fastener Tapes, Hook and Pile, Synthetic
MIL-C-29127	-	Cloth, Twill, Polyester/Cotton (Crease-Resistant Finish)
MIL-T-43548	-	Thread, Polyester, Cotton-Covered and Rayon Covered
MIL-C-44050	-	Cloth, Ballistic, Aramid

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample in which case this document shall govern.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.3 Materials. Materials shall be as specified herein (see 6.5). For materials for which it is stated "or equal", if the contractor proposes to use a material considered to be equal to the material specified, prior to its use the contractor shall furnish a sample of the material, with supporting technical data to the contracting officer for subsequent evaluation by the responsible military agency.

3.3.1 Cloth, ballistic, aramid, water-repellent. The cloth for the ballistic inserts shall conform to type IA, class 2, color natural of MIL-C-44050, except that in addition to meeting the ballistic requirements specified, the finished cloth in 18 ply panels shall defeat the .357 Mag., 158 grain, JSP bullets at 1250 $\pm$ 50 fps and 9 mm, 124 grain, FMJ bullets at 1090 $\pm$ 50 fps. The cloth being tested shall be tested individually for each type of projectile. The sample unit for the cloth shall be increased as necessary to perform the required tests.

3.3.2 Tape, binding. The tape for binding the ballistic inserts shall be of 45° bias cut polyester/cotton 50/50, 3.5 oz/sq yd, 76 by 60 minimum texture, 1-7/8 $\pm$ 1/16 inches wide, white or bleached, conforming to Atlantic Bias Products Style "Polyfrost" or equal (see 4.4.1.1).

3.3.3 Cloth, twill, polyester/cotton. The cloth for the carrier shall be polyester/cotton twill, color white, conforming to class 2 of MIL-C-29127.

3.3.4 Fastener tape, hook and pile. The hook and pile fastener tapes shall conform to type I or II, class 1, color white 3050 of MIL-F-21840. The widths shall be as specified in table I.

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3.3.5 Webbing, elastic. The elastic webbing for the side closure straps shall be 1-1/2 inches wide polyester, color white, conforming to George C. Moore Co. quality 552 or equal (see 4.4.1.1).

3.3.6 Thread, polyester, cotton or rayon covered. The thread for all stitching shall be cotton or rayon covered polyester, color white, conforming to MIL-T-43548.

3.3.7 Size, identification and instruction labels.

3.3.7.1 Ballistic inserts. Each ballistic insert shall be furnished with a combination size, identification and instruction label conforming to type VI, class 14 of DDD-L-20. The contents of the size and identification label shall be combined. The item description shall be "Ballistic Insert, Front, MP-1" or "Ballistic Insert, Back, MP-1" as applicable. The label shall show fastness to accelerated laundering. Format and contents of the instruction label shall be as follows:

PROTECTIVE LEVEL

This insert provides penetration protection only against the .22 LRHV, 40 gr., RN, lead, (1050 ± 50 fps); .38 Spec., 158 gr., RN, lead, (850 ± 50 fps); 12 gauge No. 4 shot and lesser threats, When a 2nd insert is placed over this one, the two inserts provide penetration protection against the .357 Mag., 158 gr., JSP, (1250 ± 50 fps); 9 mm, 124 gr., FMJ, (1090 ± 50 fps) and lesser threats.

WASHING INSTRUCTIONS

The ballistic insert must be washed separately from the carrier and as seldom as possible. Wash in strict accordance with the following instructions:

1. MACHINE WASH at approximately 140°F (Hot, Regular Setting) using any home laundry mild detergent which contains no bleach. DO NOT USE BLEACH OR STARCH.
2. MACHINE RINSE OR HAND RINSE (3 rinses minimum) in warm water to remove all traces of soap.
3. MACHINE DRY at highest temperature setting of dryer. Dry in home dryer (140°F) for 75 minutes or dry in laundromat dryer (160° to 180°F) for 45 minutes. DO NOT DRY IN SUNLIGHT.
4. DRY THOROUGHLY BEFORE WEARING.
5. DO NOT SEND OUT TO COMMERCIAL LAUNDRY.
6. DO NOT SCRUB WITH BRISTLE BRUSH.
7. DO NOT DRY CLEAN.

3.3.7.2 Carrier. The carrier shall be furnished with a combination size, identification and instruction label conforming to type VI, class 14 of

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DDD-L-20. The contents of the size and identification label shall be combined. The item description shall be "Body Armor, Small Arms (Hand Gun) Protective, Undergarment, MP-1". The label shall show fastness to accelerated laundering. Format and contents of the instruction label shall be as follows:

PROTECTION LEVEL

This ballistic protective undergarment was designed to provide penetration protection only against the following threats:

With 1-front insert and 1-back insert - .22 LRHV, 40 gr., RN, lead, (1050 ± 40 fps); .38 Spec., 158 gr., RN, lead, (850 ± 50 fps); 12 gauge No. 4 shot and lesser threats.

With 2-front inserts and 2-back inserts - .357 Mag., 158 gr., JSP, (1250 ± 50 fps); 9 mm, 124 gr., FMJ, (1090 ± 50 fps) and lesser threats.

WASHING INSTRUCTIONS

REMOVE BALLISTIC INSERTS BEFORE WASHING CARRIER

The carrier may be washed and dried by any conventional method. The hook fastener tape should be attached to the pile fastener tape before washing.

3.4 Design. The body armor shall consist of a carrier and two removable front and two removable back ballistic inserts. Each ballistic insert shall have 9 plies of material. The carrier shall be the over-the-head type with no adjustment at the shoulders. The carrier shall form front and back pockets, shaped to firmly support the inserts and shall have four (two at each side) adjustment straps located between the waist and chest levels. The detailed construction shall be as specified in 3.5.

3.5 Construction. The construction shall conform to the requirements specified in table I and herein. Figure 1 is furnished solely for guidance and information. Should variation from the written document appear in figure 1, the written document shall govern. During the cutting and fabrication process the tolerances allowed in table I shall be utilized in such a manner that on the finished body armor, the ballistic inserts lay flat within the carrier without any bulging of the carrier at any point or with no more than 1/4 inch of clearance between the carrier and the ballistic insert around the periphery of the inserts.

3.5.1 Stitches, seams, and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams occur.

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3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1/2 inch minimum except when caught in other stitching or turned under in a hem. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.5.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area, continue over the defective area and continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs the ends of the stitching are not required to be backstitched.

3.5.1.2 Types 502, 503, 504, 505, 515, 516, and 519 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.5.1.1.1a and 3.5.1.1.1b except substitute 3/4 inch for 1 inch wherever 1 inch appears.

3.5.1.3 Automatic stitching. Automatic machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met; and at least three or more tying, overlapping or back stitches are used to secure the ends of the stitching.

3.5.1.4 Thread ends. All thread ends shall be trimmed to a length of not more than 1/4 inch.

3.5.2 Repairs. Except as otherwise specified herein, repairs are not allowed to be made to the body armor.

3.5.3 Replacement of defective components. During the spreading, cutting and manufacturing process, components of the carrier having material defects or damages that are classified as defects in 4.4.3.1 shall be removed from production and replaced with non-defective and properly matched components.

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3.6 Patterns. Standard patterns, which provide an allowance for all seams will be furnished by the Government and shall be used by the contractor as a guide for cutting working patterns only. The standard patterns shall not be altered in any way and the working patterns shall be cut to conform exactly with the standard patterns.

3.6.1 List of pattern parts and applicable materials.

3.6.1.1 Cloth, ballistic, aramid.

Item	Nomenclature	Cut
1	Ballistic, Insert, Front	18
2	Ballistic, Insert, Back	18

3.6.1.2 Cloth, twill, polyester/cotton.

Item	Nomenclature	Cut
1	Carrier, Outer, Front	1
2	Carrier, Inner, Front	1
3	Carrier, Outer, Back	1
4	Carrier, Inner, Back	1

3.7 Manufacturing operations requirements. The body armor shall be manufactured by and with the use of all operations specified in table I. The contractor is not required to follow the exact sequence of operations.

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TABLE I
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting.						
	a. The body armor components shall be cut and marked in strict accordance with patterns furnished which show size, cutting directional lines, placement of components and notches for proper assembling of all parts. A cutting tolerance of ± 1/8 inch is permitted for all parts covered by a pattern.						
	b. The fabric components visible on the exterior of the body armor shall not be drilled for positioning any part except where a component part, when attached, will completely and permanently cover the drill hole. The use of drilled holes on the ballistic material is prohibited.						
2.	Make carrier front inner shell.						
	a. Overedge along bottom edge of carrier front inner shell using 3/16 to 1/4 inch gauge.	502, 503, 504 or 505	EFd-1	6-10	70	70	
	b. Cut 3/4 inch wide pile fastener tape for bottom edge of shell in the length specified below, ± 1/8 inch, for the applicable size: - Size small regular and long - 13 inches - Size medium regular and long - 15 inches - Size large regular and long - 17 inches - Size extra large regular and long - 19 inches						

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NO.	TABLE I (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Make carrier front inner shell. (cont'd)						
	c. With outer surface of front inner shell facing up turn bottom edge under along foldline on pattern, +1/8 -0 inch. Position the applicable length pile fastener tape along bottom edge with ends located as shown on pattern, ± 1/8 inch. Stitch fold back down and fastener tape to shell with double row of stitching across width of shell. Stitch gauge shall be 5/8 ± 1/16 inch with stitching evenly located, within 1/16 inch, across width of fastener tape.	301		8-10	50	50	
3.	Make carrier front outer shell.						
	a. Cut 3/4 inch wide hook fastener tape in the same length for the applicable size as specified in operation 2b for the pile fastener tape for the front inner shell.						
	b. With inner surface of front outer shell facing up position hook fastener tape on shell in location shown on pattern, ± 1/8 inch, and stitch along sides and ends with row of stitching located 1/16 to 1/8 inch from edges.	301		8-10	50	50	
	NOTE: As an option fastener tape may be stitched to shell with a double needle machine with stitch gauge 5/8 ± 1/16 inch. Stitching shall extend across entire width of shell and shall be evenly located, within 1/16 inch, across the width of the fastener tape.						

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NO.	TABLE I (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Make carrier front outer shell. (cont'd)						
	c. Cut four lengths of 1-1/2 inches wide pile fastener tape 7-1/2 ± 1/8 inches long. Attach the four tapes on outer surface of front outer shell in location shown on pattern, ± 1/8 inch, with row of stitching along sides and ends. Stitch margin 1/16 to 1/8 inch.	301		8-10	50	50	
	d. Overedge bottom portion of front outer shell starting at one side edge adjacent to the bottom edge of the hook fastener tape and continuing around bottom of shell to the opposite side edge.	502, 503 504 or 505	EFd-1	6-10	70	70	
	NOTE: As an option the overedging may be performed simultaneously with the joining operation in operation 4 by continuing the stitching around the bottom of the shell.						
4.	Join inner and outer shells of front carrier.						
	With inside surfaces facing out, position front inner shell on front outer shell with top and side edges evenly aligned. Fold bottom edge of inner shell back along fold line shown on pattern, ± 1/8 inch. Join the inner and outer shells with stitching along the sides and neck area leaving the tips at the shoulder open. Stitch gauge for overedge stitching shall be 3/16 to 1/4 inch and stitch margin for chain stitching shall be 3/8 ± 1/16 inch. Turn the carrier right side out and work out edges of seam fully.	515, 516, or 519	SSa-2	6-10	50 (chain stitching) 70 (overedge stitching)	70 70	

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NO.	TABLE I (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make carrier back inner shell and attach label.						
	a. Cut 3/4 inch wide pile fastener tape for bottom edge of shell in the length specified below, ± 1/8 inch, for the applicable size:						
	- Size small regular and long - 14 inches - Size medium regular and long - 16 inches - Size large regular and long - 18 inches - Size extra large regular and long - 20 inches						
	b. Overedge bottom edge of back inner shell and attach the applicable length fastener tape the same as specified in operations 2a and 2c for the front inner shell.						
	c. Position the combination size, identification and instruction label on the outside surface of the back inner shell in the approximate center of the back, 3-1/2 ± 1/4 inches above the finished bottom edge and stitch on all four sides, 1/16 to 1/8 inch from edges. Label shall read from top to bottom of shell.	301		8-10	50	50	
6.	Make carrier back outer shell.						
	a. Cut 3/4 inch wide hook fastener tape in the same length for the applicable size as specified in operation 5a for the pile fastener tape for the back inner shell.						
	b. Attach hook fastener tape and overedge bottom edge of back outer shell the same as specified in operations 3b and 3d for the front outer shell.						

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NO.	TABLE I (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make four elastic side closure straps for back carrier.						
	Cut four lengths of 1-1/2 inches wide elastic webbing and four lengths of 1-1/2 inches wide hook fastener tape 3-1/2 ± 1/8 inches long. Make four straps as follows: Fold one end of elastic webbing back 3/8 ± 1/16 inch and overlap end of hook fastener tape over the fold back 1/2 ± 1/16 inch with side edges evenly aligned and hooks of fastener tape facing up. Stitch through overlap with a 1-3/8 ± 1/16 inches by 3/8 ± 1/16 inch box pattern centered on the overlap.	301		6-8	50	50	
8.	Join inner and outer shells of back carrier and catch side closure straps.						
	With inside surfaces facing out, position inner shell on outer shell with top and side edges evenly aligned. Fold bottom edge of inner shell back along fold line shown on pattern, ± 1/8 inch. Join the inner and outer shells with stitching around the sides and neck area leaving the tips at the shoulder open, During the stitching operation the four elastic side closure straps shall be caught in the stitching. Each strap shall be positioned between the inner and outer shells in the locations shown on the pattern, ± 1/8 inch. The hook side of the fastener tape shall face the inner shell and the ends of the tape shall be even with the edges of the shells. Stitch gauge for overedge stitching shall be 3/16 to 1/4 inch and stitch margin for chain stitching shall be 3/8 ± 1/16 inch.	515, 516, or 519	SSa-2	6-10	50 (chain stitching) 70 (overedge stitching)	70 70	

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NO.	TABLE I (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Join front and back carrier shells and top stitch.						
	a. With back carrier wrong side out and front carrier right side out insert front carrier within the back carrier and evenly align the edges of the shells at the tops of the shoulders. Inner shell of front carrier shall face inner shell of back carrier. Join the shells at the shoulder tops with row of stitching.	515, 516, or 519	SSa-2	6-10	50 (chain stitching) 70 (overedge stitching)	70 70	
	b. Turn back carrier right side out and work out all seams fully around the outer periphery, shoulder tips and neck. Top stitch around neck and entire outer periphery of carrier including bottoms of the outer shells. Stitch margin shall be 3/16 ± 1/16 inch. At the bottoms of the outer shells fold edges back 3/8 ± 1/16 inch.	301		8-10	50	50	
10.	Make front and back ballistic inserts.						
	NOTE: Two individual front ballistic inserts and two individual back ballistic inserts are required for each body armor assembly. Each individual insert shall consist of a total of 9 plies assembled as follows:						
	a. Position the combination size, identification and instruction label on the outside surface; of the first ply of each ballistic insert in the approximate center 3-1/2 ± 1/4 inches above the bottom edge and stitch along all four sides, 1/16 to 1/8 inch from edges. Label shall read from top to bottom of the insert assembly.	301		8-10	50	50	

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					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Make front and back ballistic inserts. (cont'd)						
	b. With the instruction label on the first ply facing out superimpose the 9 plies of the insert on top of each other and evenly align all edges. Simultaneously join the plies together and bind the edges with the binding tape specified in 3.3.2. Fold each edge of tape under 1/4 ± 1/16 inch and bind insert around entire periphery with single row of stitching located 1/16 to 1/8 inch from edge of binding. Ends of tape shall be overlapped 1 inch minimum and end of tape on outside shall be turned under 1/2 inch minimum.	301	BSc-1	8-10	50	50	
	or						
	c. Bind periphery of insert in two operations using seam type indicated. Align edge of binding tape even with edge of insert and stitch all around periphery using 3/8 ± 1/16 stitch margin. Turn tape over onto opposite side, turn edge of binding under 3/8 ± 1/16 inch and stitch around periphery of insert with stitching located 1/16 to 1/8 inch from edge of binding. Ends of tape shall be overlapped 1 inch minimum and end of tape on outside shall be turned under 1/2 inch minimum.	301	BSg-2	8-10	50	50	
11.	Clean armor.						
	Remove all spots, stains and ends of thread.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Install ballistic inserts in carrier. Install the front and back ballistic inserts within the carrier and secure fastener tapes. Secure side closure straps across front of carrier.						

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3.8 Workmanship. The finished body armor shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the specified quality level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the documents where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3.1 and 4.4.3.2. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Certification. The contractor shall furnish a certificate of compliance certifying that the binding tape and elastic webbing specified in 3.3.2 and 3.3.5 conform to the specified requirements.

4.4.2 In-process inspection.

4.4.2.1 Visual examination of working patterns. Examination of the working patterns shall be made to assure that they conform to the Government patterns in all respects.

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4.4.2.2 In-process visual examination of ballistic inserts. Prior to binding, each ply of all ballistic inserts shall be examined for defects. Any ply containing a defect which is classified as such in MIL-C-44050 shall be removed and replaced. In addition, all inserts shall be examined to assure that they contain nine plies.

4.4.3 In-process dimensional examination of ballistic inserts. Prior to binding, the ballistic insert plies shall be examined for the defects listed below. The lot size shall be expressed in units of front and back inserts of one size only. The sample unit shall be one front or back insert. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.0 for major defects and 10.0 for total (major and minor combined) defects.

Examine	Defect	Classification Major	Minor
Size of ballistic inserts	a. Smaller than cutting pattern:		
	(1) 3/16 inch to 3/8 inch at any point around the periphery. 1/ 2/		X
	(2) More than 3/8 inch at any point around the periphery. 1/ 2/	X	
	(3) More than 1/8 inch but not more than 3/16 inch around entire periphery. 2/		X
	(4) More than 3/16 inch around entire periphery. 2/	X	
	b. Larger than cutting pattern by 1/8 inch or more at any point. 1/ 2/		X

1/ To be scored when condition exists for a length of more than 2 inches or if condition exists in several areas with an accumulated distance of 4 inches or more.

2/ The front insert or back insert, as applicable, shall be examined with the applicable cutting pattern centered on the insert.

4.4.3 End item examination.

4.4.3.1 Visual examination. The end item (carrier with two front and two back ballistic inserts) shall be examined for the defects listed below. The lot size shall be expressed in units of end items. The sample unit shall be one end item.

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The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 15.0 for total (major and minor combined) defects.

Examine	Defect	Classification Major	Minor
Cloth, carrier	a. Any hole (except drill hole), cut, or tear.	X	
	b. Any exposed drill hole.		X
	c. Any abrasion marks, broken or missing yarns, or multiple floats.	X	
	d. Any mend, darn, or patch.	X	
	e. Not cut in accordance with pattern with respect to material direct-ional requirements.		X
	f. Any needle chew.	X	
Cloth, binding	a. Any hole, cut, or tear.	X	
	b. Not cut on a bias.	X	
Fastener tape	a. Cut or tear.	X	
	b. Hooks crushed affecting closure.	X	
Webbing, elastic	a. Any hole, cut, tear, or frayed edge.	X	
	b. Needle chew: -Up to and including 1/8 inch in length. -More than 1/8 inch in length.	 X	X
Seams and stitchings:			
Open seams	a. Up to and including 1/2 inch.		X
	b. More than 1/2 inch.	X	
	NOTE: A seam shall be classified as an open seam when one or more stitches joining a seam are broken, or when two or more consecutive skipped or runoff stitches occur.		

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Examine	Defect	Classification Major	Classification Minor
Seams and stitchings: (cont'd)			
Raw edges (except where required or otherwise classified herein)	More than 1/4 inch when securely caught in stitching.		X
	NOTE: Raw edges not securely caught in stitching shall be classified as open seams.		
Seam and stitch type	a. Wrong seam or stitch type.	X	
	b. Seam pleated or excessively puckered, clearly noticeable.		X
Stitch tension	a. Tension loose, resulting in loose bobbin or top thread.		X
	b. Tension excessively tight, resulting in puckering of material.		X
Stitches per inch (all types of stitching)	a. One stitch less than minimum specified. 1/		X
	b. Two or more stitches less than minimum specified. 1/	X	
	c. Two or more stitches in excess of maximum specified. 1/		X
	NOTE: Variation in the number of stitches per inch caused by operator speeding up the machine and pulling the material in order to sew over heavy places, or in turning corners shall be classified as follows: a. Within the minor defect classification - no defect. b. Within the major defect classification - minor defect.		

1/ Score only where the condition exists for 3 inches or more or in several areas with an accumulated distance of 5 inches or more in an individual seam.

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Examine	Defect	Classification Major	Minor
Seams and stitchings: (cont'd)			
Rows of stitching	Any omitted.	X	
Thread breaks, skipped stitches or run-offs (on type 301 stitching)	Overstitched less than 1 inch in each direction beyond defective area.		X
Thread breaks, skipped stitches or run-offs (on overedge stitching)	Overstitched less than 3/4 inch in each direction beyond the defective stitching area.		X
	NOTE: On all types of stitching, thread breaks or two or more consecutive skipped or run-off stitches not overstitched shall be classified as open seams.		
Stitching ends (on type 301 stitching)	Ends of stitching not secured as specified (except when caught in other stitching or turned under in a hem).		X
Assembly - general	a. Any fastener tape reversed, i.e., pile tape attached where hook tape is required or vice versa.		X
	b. End of side closure strap not fully caught in carrier joining seam.	X	
	c. Side closure strap attached with hook side of fastener tape facing wrong direction.	X	
	d. Binding tape loosely applied: -but not exposing raw edge of material.		X
	-exposing raw edge of material.	X	
	e. End of binding tape not turned under.		X

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Examine	Defect	Classification Major	Minor
Fit of ballistic inserts within carrier	a. Carrier smaller than ballistic insert(s), i.e. not permitting assembly to lie smooth and flat within.	X	
	b. Carrier larger than ballistic insert(s) by more than 1/4 inch.		X
Components and assembly	a. Any component caught in any unrelated operation of stitching (not otherwise classified herein).	X	
	b. Any component part omitted or not attached as specified (not otherwise classified herein).	X	
	c. Any unauthorized repair.	X	
Cleanness	Any spot or stain clearly noticeable; thread ends not trimmed throughout as specified.		X
Identification	Omitted, incorrect, illegible, or not as specified.		X

4.4.3.2 Dimensional examination. The end item (carrier with two front and two back ballistic inserts) shall be examined for compliance with all dimensions specified in table I including stitch margins and gauge. Any dimension exceeding the applicable tolerance shall be classified as a defect. The lot size shall be expressed in units of end items. The sample unit shall be one end item. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 10.0.

4.4.4 Packaging inspection. An examination shall be made to determine that the preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Examination of closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of bundles per shipping container is more or less than specified.

4.4.5 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each complete body armor, with back panel superimposed over the front panel so that the identification label is visible and with the carrier cloth bottom folded under, shall be inserted flat in a close-fitting clear polyethylene film bag having a thickness of 0.0015 inch (± 20 percent tolerance). The bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the heat seal made as close as possible to the open end. The bag

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may be fabricated from polyethylene film tubing or sheeting. A 1/4-inch diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape. Alternatively, the final closure of the polyethylene bag may be accomplished by means of a tuck or reverse flap.

5.1.2 Commercial. Each complete body armor shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Ten complete body armors, of one size only, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The body armor shall be packed flat, 10 in depth, each alternatively reversed top to bottom, within a shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636 except the inspection shall be in accordance with 4.4.4. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2 Level B packing. Ten complete body armors, of one size only, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. The body armor shall be packed flat, 10 in depth, each alternately reversed top to bottom, within a shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636 except the inspection shall be in accordance with 4.4.4. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Body armor, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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5.3 Palletization. When specified (see 6.2), body armor packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The body armor is intended for use by military police for protection against small arms (hand gun) fire. (See 3.3.7.2 for levels of protection.)

6.1.1 Application of sizes. The body armor sizes are intended to fit the following chest and height measurements:

Size	Chest measurement (inches)	Height measurement
Small regular	34-38	Up to 71 inches
Medium regular	38-42	Up to 71 inches
Large regular	42-46	Up to 71 inches
Extra large regular	46-50	Up to 71 inches
Small long	34-38	71 inches or more
Medium long	38-42	71 inches or more
Large long	42-46	71 inches or more
Extra large long	46-50	71 inches or more

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document
b. When a first article sample is required (see 3.2).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. Whether weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
e. When palletization is required (see 5.3).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

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6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample consisting of one finished body armor assembly. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document.

6.6 Supersession data. This document includes the requirements of Limited Production Purchase Description 1-78A, 21 April 1978, Body Armor, Small Arms, (Hand Gun) Protective, Undergarment, MP-1.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 82

User activity:

Air Force - 45

Preparing activity:

Army - GL

Project No. 8470-0101

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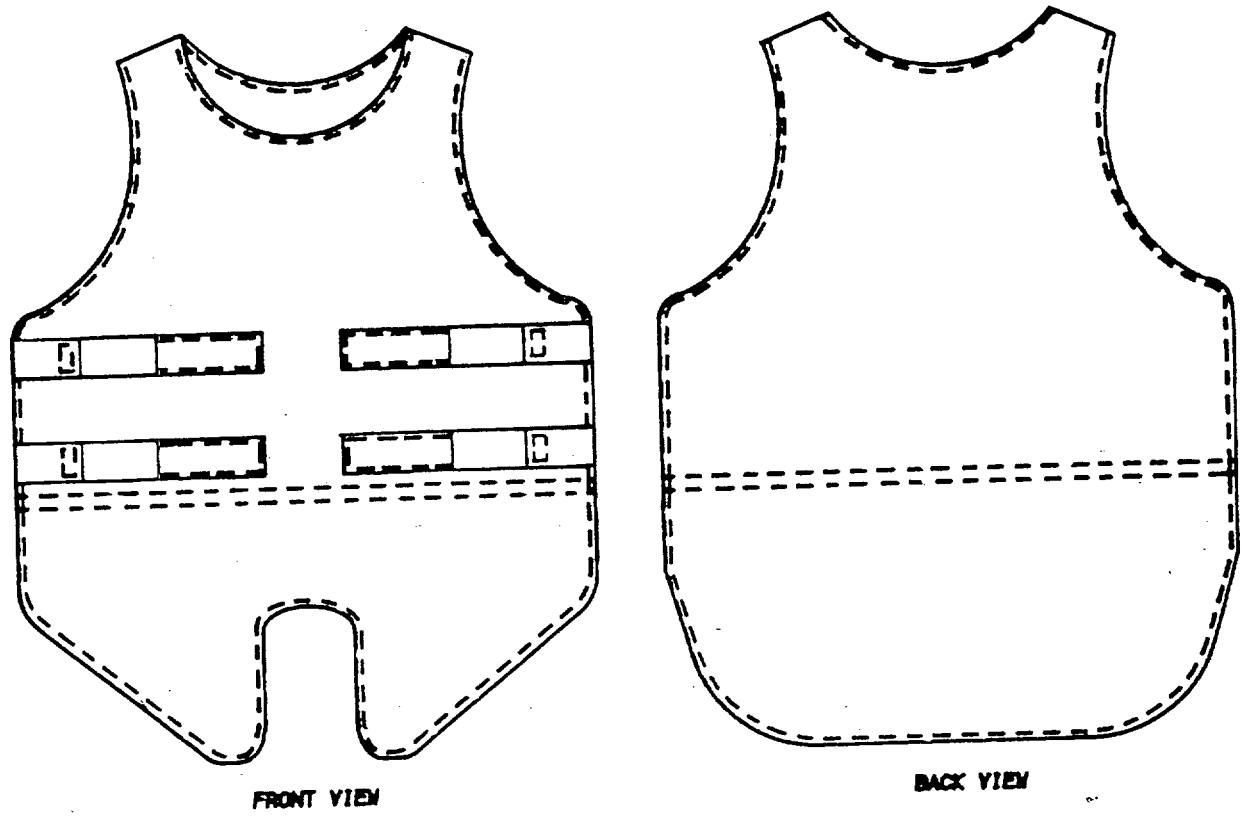


FIG-1: BODY ARMOR, SMALL ARMS, (HAND GUN) PROTECTIVE, UNDERGARMENT, MP-1.

2-1-1627

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